



SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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SIMATS
ENGINEERING

Digital Library & Paperless Learning Resource Management System

**OBJECT ORIENTED PROGRAMMING WITH C++
DSA- 0101**

GOKUL K
1911260026

GOWTHAM.S
1971260007

SIDHARDHA .V
1965260034

Faculty:Dr.k. Parimala Kanaga Devan | SLOT D|July- sep 2026

Abstract:

The Digital Library & Paperless Learning Resource Management System is a C++-based application designed to manage and access educational resources digitally. It provides a centralized platform for storing, organizing, searching, and retrieving resources such as e-books, notes, question papers, study materials, and research papers. The main objective is to reduce the use of physical documents and promote a paperless learning environment. Students can easily search and access required learning materials, while administrators can add, update, delete, and manage resources efficiently. The system uses Object-Oriented Programming concepts in C++, including classes, objects, encapsulation, inheritance, polymorphism, and file handling. It improves the accessibility, efficiency, and organization of academic resources while reducing paperwork and manual record maintenance.

Literature Survey (2024–2025)

Recent innovations, gaps identified, and our proposed augmentations.

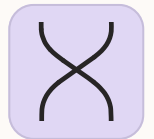
Author	Year	Method	Pros / Cons	Our Solution
Lee	2025	Offline-first reader app	Pros: accessibility. Cons: sync conflicts	Conflict-resilient sync with versioning
Nguyen & Silva	2025	Open-access analytics	Pros: transparency. Cons: privacy concerns	Privacy-preserving aggregated analytics
Kumar et al.	2024	AI-based content tagging	Pros: faster indexing. Cons: bias in models	Human-in-loop validation + fairness checks

Literature Survey (2023–2024)

Comparative overview of recent work: methods, strengths, weaknesses, and how our approach builds on them.

Author	Year	Method	Pros / Cons	Our Solution
Rodríguez	2024	Embedded micro-LMS	Pros: seamless course flow. Cons: limited analytics	Modular LMS bridge + advanced usage analytics
Smith et al.	2023	Metadata-first repository	Pros: rich discovery. Cons: heavy curation load	Automated metadata enrichment + curator dashboard
Wang & Patel	2023	Decentralised access control	Pros: scalable. Cons: complex key management	Hybrid RBAC with SSO and audit trails

Summary of Cons Found in Literature



High curation effort

Manual tagging and quality checks slow scaling.



Complex access control

Key management and policy enforcement are brittle.



Poor analytics

Limited insight into resource impact and learning outcomes.



Offline & sync issues

Readers struggle with reliable sync and conflict resolution.

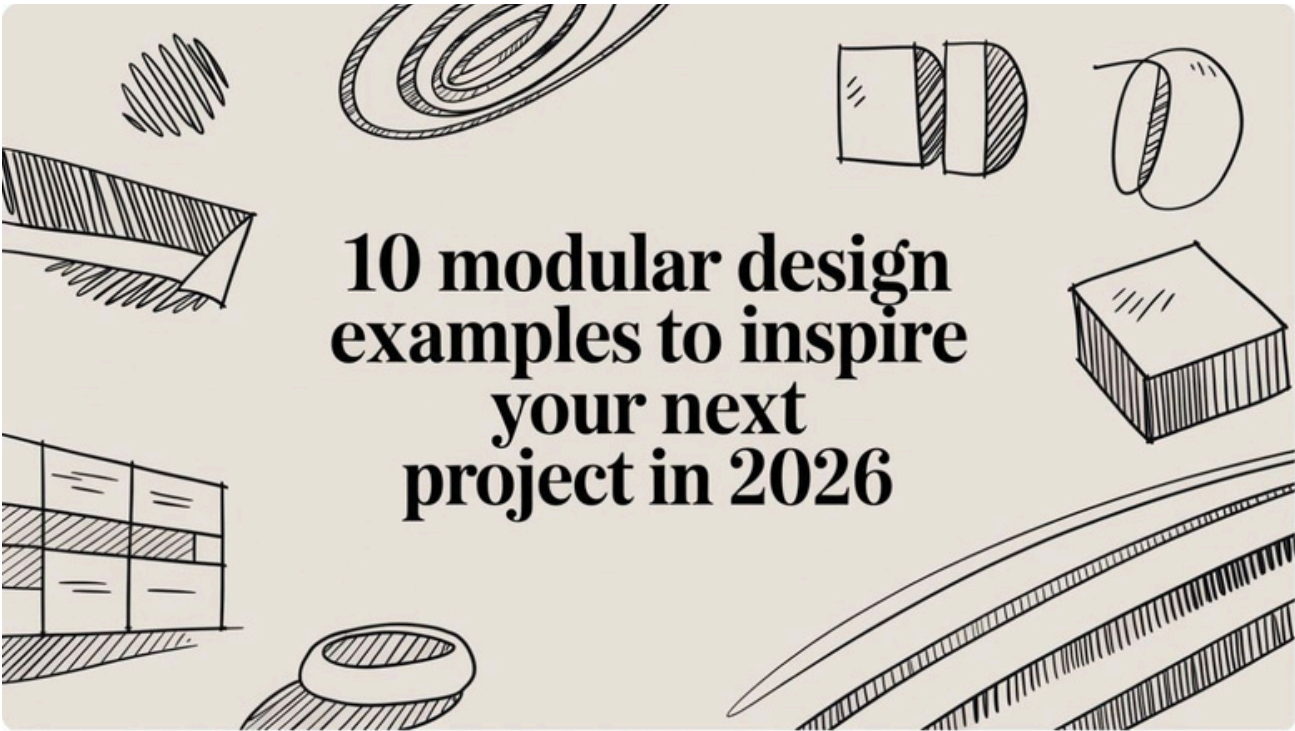


We designed our system specifically to address these four recurring limitations.

Aim of the Project

To deliver a secure, user-centred digital library and resource-management platform that: enhances discoverability, reduces manual overhead, supports paperless workflows, integrates with course systems, preserves privacy, and provides actionable analytics for educators and librarians.

System Modules



Discovery Engine

Advanced search, semantic metadata enrichment, recommendations.



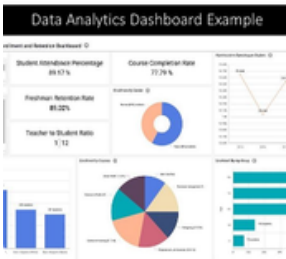
Access & Rights

SSO, RBAC, licence tracking, fair-use handling.



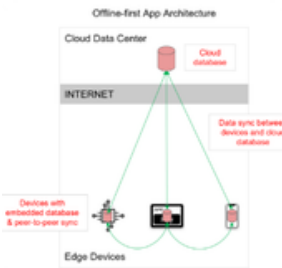
Learning Integrations

Lightweight LMS bridge, assignment linking, inline reading & annotation.



Analytics & Reporting

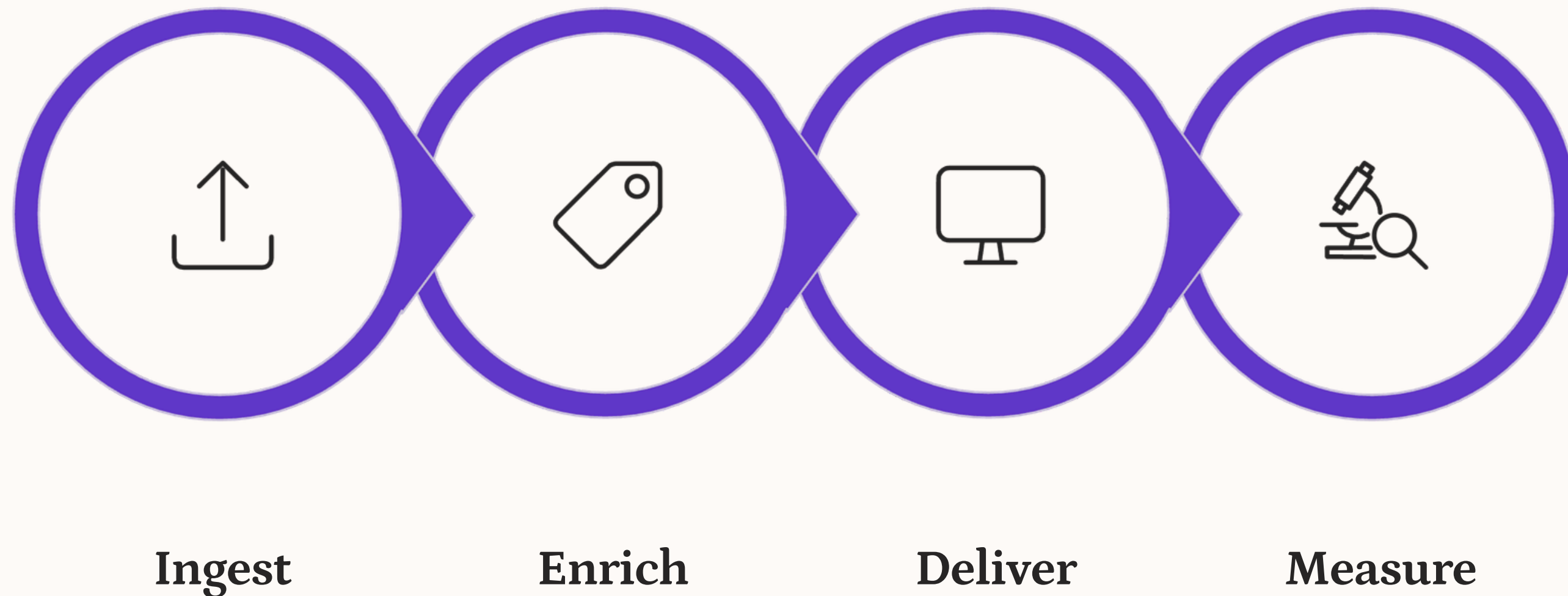
Privacy-preserving usage metrics, content impact reports.



Offline Reader & Sync

Conflict-resilient sync, versioning, compressed local storage.

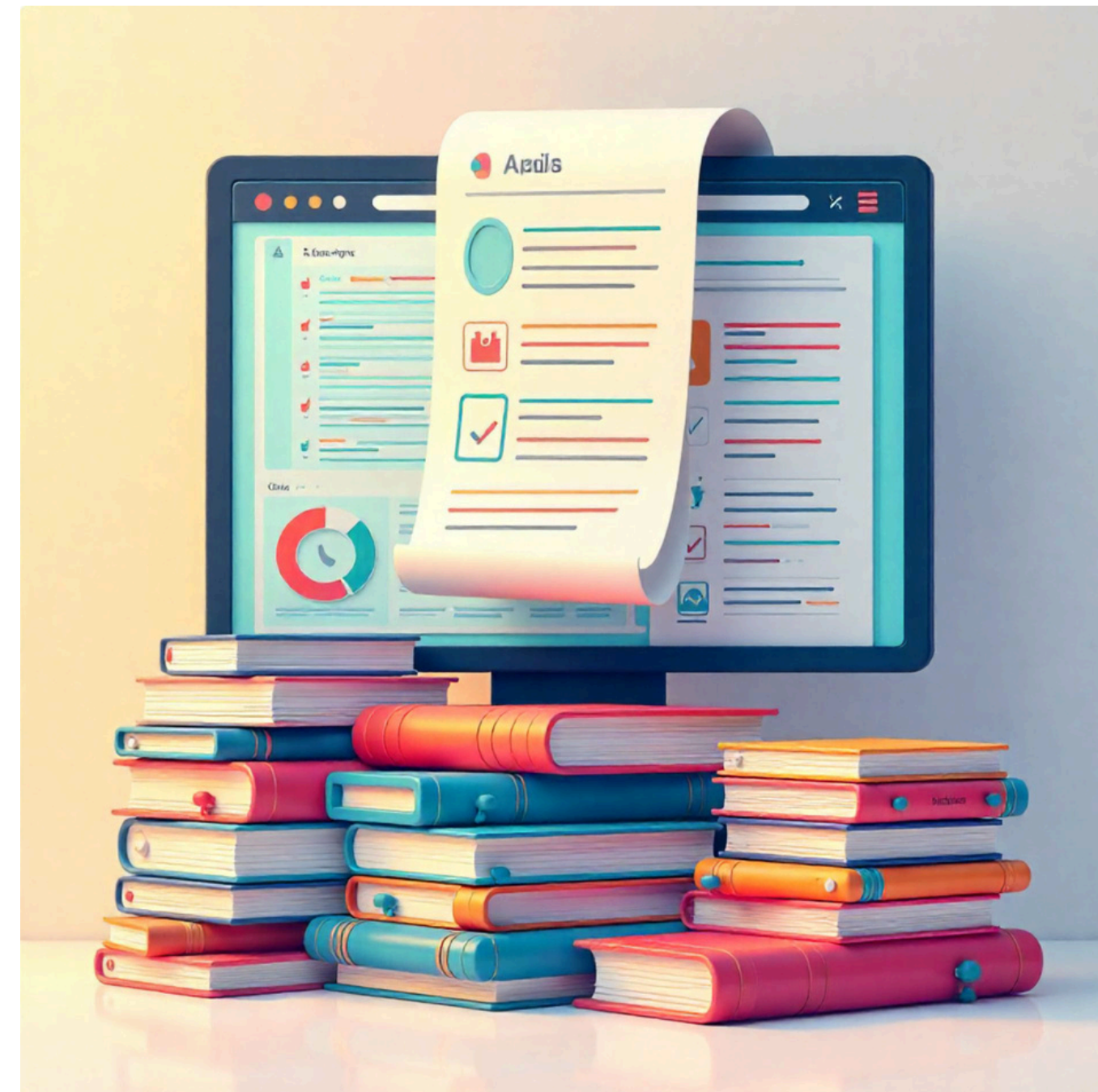
Conclusion



Outcome: A practical roadmap to transition institutions to paperless learning while improving discovery, reducing labour, and measuring educational impact. Next steps: prototype, pilot with two faculties, iterate based on analytics and user feedback.

References

- Smith, J., et al. (2023). Metadata-first repositories.
- Wang, L., & Patel, R. (2023). Decentralised access for digital libraries.
- Kumar, A., et al. (2024). AI-assisted tagging for education.
- Lee, H. (2025). Offline-first educational readers.



Thank you

